

TIKHORETSK, Russian Federation

WMO: 348380

Lat: 45.850N

Lon: 40.083E

Elev: 78

StdP: 100.40

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.9	-12.3	-19.0	0.7	-15.0	-15.7	1.0	-11.8	8.3	1.4	7.3	-0.1	2.6	10	0.441

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	35.1	21.6	33.3	20.8	31.5	20.2	23.5	30.9	22.4	29.8	21.6	28.8	2.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.2	16.0	26.9	20.1	14.9	25.8	19.2	14.1	24.9	70.6	31.1	66.3	29.6	63.2	28.8	31.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.2	5.4	-18.6	37.4	5.6	1.5	-22.6	38.5	-25.9	39.4	-29.0	40.3	-33.1	41.4		
(5)			-19.3	25.9	5.3	2.6	-23.1	27.8	-26.2	29.4	-29.2	30.9	-33.1	32.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	-1.1	0.1	5.1	11.7	17.6	21.6	24.7	24.2	18.7	11.8	5.3	1.0		
	DBStd	10.18	5.85	5.75	4.35	4.09	4.04	3.30	3.04	3.50	3.73	4.93	4.88	5.29		
	HDD10.0	1282	344	277	161	27	1	0	0	0	1	37	152	281		
	HDD18.3	2977	602	510	411	202	64	8	0	2	40	211	390	538		
	CDD10.0	1933	0	1	9	78	236	347	456	441	261	92	12	1		
	CDD18.3	587	0	0	0	3	41	105	198	184	50	7	0	0		
	CDH23.3	5809	0	0	1	33	369	923	2087	1955	388	53	0	0		
	CDH26.7	2357	0	0	0	4	93	322	933	892	102	12	0	0		
(14) Wind	WSAvg	2.5	2.8	2.9	3.1	2.9	2.3	2.1	2.1	2.1	2.2	2.3	2.6	2.7		
(15) Precipitation	PrecAvg	622	53	36	39	47	63	77	59	45	40	40	55	65		
	PrecMax	930	139	123	83	136	168	218	135	116	79	88	118	181		
	PrecMin	414	4	6	0	1	12	0	2	3	1	2	4	8		
	PrecStd	102	34	24	21	30	32	48	33	33	21	21	34	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	15.9	20.7	26.8	31.8	34.2	37.0	37.8	32.4	28.0	20.1	14.1		
		MCWB	7.9	10.1	11.5	15.0	19.4	21.3	22.7	21.3	21.4	17.4	12.9	9.9		
	2%	DB	9.0	11.5	16.8	23.5	29.2	32.2	35.2	35.3	29.4	24.2	16.2	11.0		
		MCWB	6.5	7.6	9.8	14.0	18.1	20.4	21.9	21.5	19.2	16.3	12.1	8.1		
	5%	DB	7.0	9.2	14.0	21.1	27.3	30.2	33.4	33.4	27.4	21.4	13.9	9.1		
		MCWB	5.0	6.3	8.7	12.9	17.5	19.7	21.1	20.7	18.6	15.2	10.9	7.0		
	10%	DB	5.4	7.2	11.6	19.0	25.4	28.3	31.5	31.3	25.5	18.8	11.7	7.5		
		MCWB	3.9	4.9	7.5	11.9	16.7	19.3	20.5	20.1	17.6	14.1	9.4	5.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.7	10.3	12.6	16.2	20.9	24.1	25.7	25.1	23.2	18.2	13.8	10.6		
		MCDB	10.5	14.6	18.3	23.3	28.8	29.5	33.2	31.3	30.6	25.3	17.5	13.1		
	2%	WB	6.7	8.0	10.7	14.8	19.6	22.2	23.7	23.2	20.7	17.0	12.5	8.5		
		MCDB	8.6	11.2	15.8	21.8	26.9	29.3	31.4	30.8	27.3	23.2	15.9	10.7		
	5%	WB	5.2	6.5	9.2	13.7	18.5	21.3	22.5	22.2	19.5	15.8	11.1	7.2		
		MCDB	6.9	9.1	13.2	19.5	25.2	28.0	29.9	29.8	25.2	20.2	13.6	9.0		
	10%	WB	3.9	5.0	7.8	12.7	17.5	20.2	21.6	21.3	18.5	14.6	9.6	5.9		
		MCDB	5.3	7.1	11.2	17.7	23.6	26.3	28.9	28.6	23.8	18.3	11.4	7.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.7	6.8	7.9	9.8	10.2	10.1	10.8	11.3	10.6	8.8	6.7	5.5		
		MCDBR	7.4	9.7	12.5	13.3	12.7	12.3	12.7	13.4	13.0	12.7	10.3	7.8		
		MCWBR	5.6	6.5	7.1	6.2	5.5	4.9	5.1	5.4	6.0	6.4	6.5	5.9		
		MCWBR	7.2	9.2	11.1	11.3	11.1	11.2	11.2	11.5	11.5	11.4	9.4	7.4		
(40) Clear-Sky Solar Irradiance	taub	taub	0.346	0.378	0.418	0.456	0.420	0.428	0.456	0.452	0.420	0.389	0.355	0.348		
		taud	2.262	2.179	2.117	2.061	2.251	2.260	2.193	2.201	2.266	2.311	2.340	2.275		
		Ebn at Noon	729	774	797	802	849	842	812	798	790	758	716	679		
		Edh at Noon	84	110	135	156	133	133	140	134	115	96	77	75		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.19	2.07	3.05	4.70	5.93	6.45	6.45	5.80	4.25	2.63	1.42	0.97		
		RadStd	0.20	0.30	0.27	0.46	0.60	0.40	0.31	0.38	0.41	0.32	0.19	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	+0.48	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+99	+60

Nomenclature: See separate page